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SUPERSEDING
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December 28, 1975

FEDERAL SPECIFICATION

SCALE REMOVING COMPOUND (SOLID AND LIQUID FORM)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers two types of scale removing compounds (see 6.1).

1.2 Classification.

1.2.1 Types. The scale removing compounds covered by this specification shall be of the following types, as specified (see 6.2).

Type I - Powder, sulfamic acid.
Type II - Liquid, phosphoric acid.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals.
PPP-D-723 - Drums, Fiber.

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

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(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

16 CFR 1500 - Federal Hazardous Substances Act.
49 CFR 170-189- Hazardous Materials Regulations.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D 460 - Sampling and Chemical Analysis of Soaps and Soap Products.
D 1331 - Surface and Interfacial Tension of Solutions of Surface-Active Agents.
D 1429 - Specific Gravity of Water and Waste Water.
D 2834 - Nonvolatile Matter (Total Solids) in Water-Emulsion Floor Polishes, Solvent-Based Floor Polishes, and Polymer-Emulsion Floor Polishes.
E 70 - pH of Aqueous Solutions with the Glass Electrode.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

United States Pharmacopeial Convention, Inc. (USP):

USP Sulfanilamide Reference Standard.

(USP Reference Standards and application for copies of the USP publications should be addressed to the United States Pharmacopeial Convention, Inc., 12601 Twinbrook Parkway, Rockville, MD 20852.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

. REQUIREMENTS

3.1 Material.

3.1.1 Type I. The scale removing compound shall be a clean, uniform, free-flowing (lump-free) powder and meet the requirements specified in 3.2 and 3.3. The finished formulation shall contain an anti-caking agent, corrosion inhibitor, and not less than 90.0 percent by weight of sulfamic acid when tested as specified in 4.3.1. In addition, the product shall contain metanil yellow dye with a dye stabilizer to indicate that the scale removing compound is spent during the descaling process.

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3.1.2 Type II. The scale removing compound shall be a uniform liquid, free of sediments and suspended particles. It shall consist of phosphoric acid in combination with various acids, corrosion inhibitor, and wetting agents, and shall meet the requirements as specified in 3.2, 3.3, and table I.

3.1.2.1 Excluded ingredient. The scale removing compound shall contain no sulfuric acid or organic solvents.

TABLE I. Chemical and physical requirements

Characteristics	Minimum	Maximum
Total acidity (as phosphoric acid), percent	30.0	-----
Phosphoric acid (as P205), percent	21.0	-----
Specific gravity at 20°C/20°C (68°F/68°F)	1.17	1.30
pH value, 1.0 percent solution	-----	2.00
Surface tension at 25°C (77°F), 1.0 percent solution, dynes per centimeter	-----	44.0
Hydrochloric acid, percent	-----	10.0
Nonvolatile matter at 105°C. (221°F), percent	30.0	-----
Corrosive effect on steel, loss of weight of steel, percent	-----	0.15

3.2 Labeling (all types). The containers shall be labeled in accordance with the Federal Hazardous Substances Act and Department of Transportation Rules and Regulations. Each container of scale removing compound of specified type shall be durably and legibly labeled with precautions and directions for use.

3.2.1 Warning label. All products shall be labeled with the following warning information:

WARNING: INJURIOUS TO SKIN AND EYES
HARMFUL IF SWALLOWED

Contains acid.
Avoid contact with eyes, mucous membranes, skin, and clothing.
In case of skin contact, flush with water.
In case of contact with clothing, remove and wash contaminated clothing before re-use.
For eye contact, flush with water for 15 minutes. Get prompt medical attention.
If swallowed give large quantities of water or milk. Call a physician.
Keep out of the reach of children.

3.2.2 Directions for use.

3.2.2.1 Type I. In addition to the manufacturer's recommended dilutions and "Directions for Use", each container of scale removing compound shall have the following information:

(a) Drain water from tanks of the commercial dishmachine, flush out dirt and sediment with hose.
(b) Refill tanks with fresh hot water under 150°F (65.5°C).
(c) Feed water line should be shut off after tanks are filled to capacity.
(d) Add descaling compound to the tank.
(e) Start circulating pumps and run equipment until evidence shows that no further descaling action is taking place or indicator shows that compound is spent.
(f) Drain tanks and inspect for evidence of scale.
(g) Repeat if needed as in (b).
(h) Flush thoroughly with fresh water until all descaling compound is removed.

3.2.2.2 Type II. The manufacturer's recommended dilutions and "Directions For Use" shall be clearly shown on the label.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated herein.

4.2.2 Inspection and test of the end item. The end item shall be examined in accordance with the classification of defects, inspection levels and acceptable quality levels (AQL's), set forth below. The lot size, for purpose of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of filled primary containers of descaling compound of the same type for examinations under 4.2.2.2.

4.2.2.1 Examination of the end item container for defects in appearance of descaling compound. The sample unit for this examination shall be one open, filled, primary container. The inspection level shall be S-2 with an AQL of 2.5 percent defective.

Examine	Defects
Appearance of scale removing compound	Form not as specified. Not free-flowing. Lumping or caking. (Type I only.) Presence of dirt or foreign matter. Not homogeneous. Any evidence of segregation. (Type II only.) Not clear. (Type II only.) Not pourable, too viscous. (Type II only.)

4.2.2.2 Examination of preparation for delivery. An examination shall be made to determine compliance with the requirements of section 5. Defects shall be scored as specified in table II. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one shipping container fully prepared for delivery. The inspection level shall be S-2 with an AQL of 4.0 percent defective.

TABLE II. Examination of preparation for delivery

Examine	Defects
Containers	Not as specified.
Contents	Not as specified.
Markings	Omitted; incorrect; illegible; improper size, location, sequence or method of application.
Materials	Components missing or damaged.
Weight	Quantity per package or shipping container less than specified.
Workmanship	Bulging or distortion of containers. Cushioning inadequate, improper or missing. Inadequate closure.

4.2.2.2.1 Examination for closure, waterproofing, and banding of shipping container. When shipping containers are required to be in accordance with PPP-B-636, examination for defects in closure, waterproofing, and banding shall be made in accordance with the appendix to that specification.

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4.2.2.3 Testing of the end item. The end item shall be tested for the characteristics listed in table III. Sixteen ounces of the scale removing compound of type I or 1 pint of the scale removing compound of type II shall be drawn from each of five shipping containers and composited. The test sample shall consist of a 32-ounce composite of type I scale removing compound or a 4-pint composite of type II scale removing compound. Unless otherwise specified, duplicate determinations shall be performed on the composite with no evidence of failure to meet the specified requirements.

4.3 Test procedures.

4.3.1 Sulfamic acid.

4.3.1.1 Reagents.

Standard sodium nitrite solution, 0.25 normal. Standardize against U.S.P. Sulfanilamide as follows:

Weigh accurately about 1.25 ± 0.01g of U.S.P. Sulfanilamide Reference Standard previously dried at 105°C for 3 hours and transfer to a beaker. Add 50 ml of distilled water and 5 ml of concentrated hydrochloric acid and stir well until dissolved. Cool to 15°C, add about 25g of crushed ice, and titrate slowly with the sodium nitrite solution, stirring vigorously, until a glass rod dipped into the titrated solution produces an immediate blue ring when touched to starch-iodide paper. When the titration is complete, the end-point should be reproducible after the mixture has been allowed to stand for 1 minute. Calculate the normality.

$$\text{Normality} = \frac{\text{G of standard}}{\text{ml of sodium nitrite solution used} \times 0.1722}$$

4.3.1.2 Apparatus.

(a) pH meter.
(b) Platinum-calomel electrodes.
(c) Buret, 50-ml capacity, graduated in 0.1-ml divisions.
(d) Volumetric flask, 500-ml capacity.
(e) Magnetic stirrer.
(f) Pipet, 100-ml capacity.

TABLE III. Instructions for testing end item

Characteristics	Specification reference		Number determination per unit composite	Results reported Numerically to nearest 1/	
	Requirement	Test method			
Sulfamic acid	3.1.1	4.3.1	2	0.1	percent
Total acidity, as phosphoric acid	Table I	4.3.2	2	0.1	percent
Phosphoric acid, as P205	Table I	4.3.3	2	0.1	percent
Specific gravity	Table I	4.3.4	2	0.0001	unit
pH value	Table I	4.3.5	2	0.1	pH unit
Surface tension	Table I	4.3.6	2	1.0	dyne/cm
Hydrochloric acid	Table I	4.3.7	2	0.1	percent
Nonvolatile matter	Table I	4.3.8	2	0.1	percent
Corrosive effect on steel	Table I	4.3.9	2	0.001	percent

1/ Test report shall include all values on which results are based.

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4.3.1.3 Procedure. Dissolve a 5 g-sample weighed to the nearest mg of the scale removing compound, in distilled water, transfer to a 500-ml volumetric flask and dilute to the mark with distilled water. Transfer a 100-ml aliquot of the solution to a 400-ml beaker. Add 20 ml hydrochloric acid, cool to about 15°C with crushed ice, and slowly titrate with 0.25 N sodium nitrite that has been standardized. Determine the end-point with a pH meter using platinum-calomel electrodes. Place the buret tip below the surface of the solution and stir the solution gently using a magnetic stirrer, without pulling a vortex of air under the surface, maintaining the temperature at about 15°C. When the titration is within 1 ml of the end-point (about the 40.0 ml-mark), add the titrant in 0.1-ml portions, allowing not less than 1 minute between additions. (The instrument needle deflects and then returns to approximately its original position until the end-point is reached.) Calculate the percentage of sulfamic acid as follows:

$$\text{Sulfamic acid, percent} \pm 0.5 = \frac{\text{ml sodium nitrite} \times \text{normality} \times 48.5}{\text{weight of sample (g)}}$$

4.3.2 Total acidity (as phosphoric acid).

4.3.2.1 Procedure. Weigh accurately, to the nearest mg, 3.0 ml of scale removing compound and transfer to a 250-ml beaker. Add about 100 ml of distilled water and titrate electrometrically to a pH 9.0 with 1.0 N standard sodium hydroxide solution.

$$\text{Total acidity, as phosphoric acid, percent} = \frac{V \times N}{W} \times 4.9$$

Where: V = volume of sodium hydroxide solution required for titration.
N = normality of the sodium hydroxide solution.
W = grams of sample used.

4.3.3 Phosphoric acid (as P2O5). Determine phosphoric acid (as P2O5) on a 5.0-g sample weighed to the nearest mg in accordance with ASTM D 460, sections 60 to 75.

4.3.4 Specific gravity. The specific gravity of the scale removing compound shall be determined in accordance with ASTM D 1429.

4.3.5 pH value. The pH shall be determined on a 1 percent (by weight) solution of the scale removing compound in accordance with ASTM E 70.

4.3.6 Surface tension. The surface tension shall be determined on a 1 percent (by weight) solution of the scale removing compound in accordance with ASTM D 1331.

4.3.7 Hydrochloric acid.

4.3.7.1 Reagents.

(a) Ferric alum indicator solution - Dissolve 8 g of ferric ammonium sulfate in distilled water to make 100 ml. Add nitric acid to bleach the brown solution.
(b) Nitrobenzene, reagent grade.
(c) Ammonium thiocyanate, standard solution (0.1 N).
(d) Silver nitrate, standard solution (0.1 N).

4.3.7.2 Procedure. Weigh accurately, to the nearest mg, 10.0 g of scale removing compound. Transfer to a 250-ml beaker containing 5 g of lampblack and 100 ml of distilled water. Boil for 20 minutes, cool, and filter the solution into a 250-ml volumetric flask rinsing the beaker with several portions of distilled water. Dilute to the mark with distilled water. Pipet 50.0 ml aliquot of this solution into a 250-ml Erlenmeyer flask containing 50.0 ml of standard 0.1 N silver nitrate solution, 3.0 ml of nitric acid, 3.0 ml nitrobenzene and 2.0 ml of ferric alum indicator solution. Mix and titrate the excess silver nitrate with standard 0.1 N ammonium thiocyanate solution. Calculate the percent of hydrochloric acid as follows:

$$\text{Hydrochloric acid, percent} = \frac{A \times 5 \times 3.65}{C}$$

Where: A = milliliters of 0.1 N silver nitrate solution consumed.
C = grams of sample used.

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4.3.8 Nonvolatile matter. The nonvolatile matter shall be determined on a 2.0 ± 0.001 g of scale removing compound in accordance with ASTM D 2834.

4.3.9 Corrosive effect on steel (type II).

4.3.9.1 Preparation of coupons. Descale 2-inch x 2-inch x 0.035-inch new coupons cut from SAE 1010 hot rolled steel by immersing in concentrated reagent grade hydrochloric acid for 15 minutes. The coupons shall then be rinsed, first with tap water, next with distilled water, and finally with reagent grade acetone. Dry the coupons at 105°C for 15 minutes. Weigh the coupons to the nearest milligram and store in a desiccator. Proceed with the test in 4.3.9.2 with coupons prepared on the same day of the test.

4.3.9.2 Procedure. Take 3 new coupons prepared as in 4.3.9.1 and place in 250-ml beakers, one coupon to a beaker. Rest the coupon on a section of 1/8-inch glass rod bent into a V shape. Add 200 ml of liquid scale removing compound to each beaker. Allow the scale removing compound to remain unagitated in contact with the metal coupon for 24 hours and maintain room temperature conditions. Remove the coupons after 24 hours, rinse two or three times with distilled water, and finally rinse with acetone, dry at 105°C for 15 minutes, and weigh again. Determine the percent weight loss by the following formula and average the percentages.

$$\text{Percent weight loss} = \frac{(\text{wt. before immersion}) - (\text{wt. after immersion}) \times 100}{\text{wt. before immersion}}$$

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or commercial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Type I. Type scale removing compound does not require packaging.

5.1.1.2 Type II. One gallon of scale removing compound shall be packaged in a high density (min 0.95) polyethylene bottle with a handle and a closure with a protective reclosable feature. Each bottle shall be secured with a Seal A closure conforming to PPP-C-186.

5.1.2 Commercial. The scale removing compound shall be packaged in unit containers normally used in commercial practice. The complete package shall be designed to protect the item against damage during shipment, handling, and storage.

5.2 Packing. Packing shall be level A or commercial, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Type I. One hundred pounds of type I scale removing compound shall be packed in a drum conforming to PPP-D-723, type I. The drum shall be provided with a polyethylene film bag which is slightly larger than the inside of the drum. The bag shall be securely closed to prevent sifting of the product. The thickness of the film shall be not less than 0.002 inch. The drum shall be closed in accordance with PPP-D-723.

5.2.1.2 Type II. Four or six 1-gallon bottles of scale removing compound packaged as specified in 5.1.1.2, shall be packed in a fiberboard shipping container conforming to PPP-B-636, style RSC-L, grade V2s. In addition, each container shall be provided with top and bottom pads fabricated from the same material as the box liner. The pads shall measure 1/4 inch less than the inside length and width of the box liner. Each shipping container shall be closed, waterproofed, and banded in accordance with the appendix to PPP-B-636.

5.2.2 Commercial. The scale removing compound shall be packed in containers to insure safe delivery at destination, to provide for safe redistribution by the initial receiving activity, and shall be acceptable by common carrier under the National Motor Freight Classification or Uniform Freight Classification and 49 CFR 170-189, Hazardous materials regulations.

5.2.3 Marking. The unit and shipping containers and unitized loads (when applicable) shall be marked in accordance with 3.2.2.1 and Fed. Std. No. 123 for civil agencies or MIL-STD-129, as applicable.

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5.3 Unitization. (Applicable to full rail car and truck load shipments only.) The scale removing compound shall be unitized for shipment in accordance with normal commerical practice. The unitized load shall not exceed 2,500 pounds in weight, 63 inches in height, 56 inches in length, and 42 inches in width.

6. NOTES

6.1 Intended use.

6.1.1 Type I. Type I scale removing compound covered by this specification is intended for removing scale deposits normally found in salt water evaporators, fresh water cooling and distributing systems, and water side scale found in heat transfer units. It can be used for removing scale deposits normally found in commercial dishwashing machines.

6.1.2 Type II. Type II scale removing compound covered by this specification is intended for removing scale deposits from environmental surfaces, commercial dishwashing machines, and boiler coils. It can be used in laboratories, food handling and processing areas, veterinary areas, and dairy plants. The product is authorized by the USDA for use in Category C, under U.S.D.A Poultry, Meat, Rabbit and Egg Products Inspection Program "Acid Cleaners For Use In All Departments".

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type required.
(c) Packaging and packing (see 5.1 and 5.2).

MILITARY INTEREST:

The Department of Defense has waived coordination of this document until further notice.

Preparing activity:

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CIVIL AGENCY COORDINATING ACTIVITIES:

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